

Direction 1: We shall prove $p \rightarrow q$ is True by a direct proof.

Let, p be True. $\therefore n$ is an odd integer. By Defn 1, there exists an integer k such that $n = 2k + 1$.

$$\therefore 5n + 6 = 5(2k + 1) + 6 = 10k + 5 + 6 = 2(5k + 5) + 1$$

It's easy to see that $(5k + 5)$ is an ~~odd~~ integer.

By Defn 1, $(5n + 6)$ is an odd integer. Therefore, q is True.

By Defn. of conditional, $p \rightarrow q$ is True.

Direction 2: We shall prove $q \rightarrow p$ is True by a proof of contraposition.

$\therefore$ We shall prove $\neg p \rightarrow \neg q$ is True.

$\neg p$: n is not odd, i.e. n is even (Defn 1)

$\neg q$: $5n + 6$ is not odd, i.e. $5n + 6$ is even (Defn. 1)

Let, $\neg p$ be True. $\therefore n$ is even. By defn. 1, $\exists$ an integer k such that

$$n = 2k \quad \therefore 5n + 6 = 5(2k) + 6 = 10k + 6 = 2(5k + 3)$$

$\therefore k$ is an integer, $\therefore 5k + 3$ is an integer. By Defn 1, $(5n + 6)$ is even.

$\therefore \neg q$ is True. By defn. of conditional, $\neg p \rightarrow \neg q$ is True.

$\therefore (q \rightarrow p)$ is True. $\therefore p \leftrightarrow q$ is True

So, n is odd if and only if $(5n + 6)$ is odd $\square$

30) Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.

Solution: $p: m^2 = n^2$ $q: m = n$ $r: m = -n$

We need to prove that $p \leftrightarrow (q \vee r)$ is True. We need to prove that $(p \rightarrow (q \vee r)) \wedge ((q \vee r) \rightarrow p)$ is True. By defn. of AND, we need to prove that both $p \rightarrow (q \vee r)$ and $(q \vee r) \rightarrow p$ are respectively True.